Week 1: Financial Asset Prices and Returns

Motivation

trade off btw risk and return

1. Much of finance is concerned with measuring and managing financial risk.
2. **Risk:** uncertainty in future returns from an investment.
3. Fin. Metrics is the study of random phenomena associated with financial risk. (mostly returns from an investment). Combines:
    - Finance Theory,
    - Empirical Finance,
    - Statistical Theory,
    - Practical applications of statistics.
4. Interests include stock prices, their movements (volatility), asset pricing.

no risk ⇒ no finance

This Course

What are the goals of this course?

- Analyze and understand the stylized facts of asset returns.
- Learn to apply econometrics methods capable of modeling financial data.
- Discuss the potential pitfalls of these econometric methods.

Simple returns

"t" $\Rightarrow$ today

"$t-1$" $\Rightarrow$ yesterday

① $P_t - P_{t-1} > 0$

$\downarrow$

compare

② $\div$ start from P_{t-1}

Simple returns $R_t > 0$

The simple return on an asset between time $t-1$ and t is given by

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

If the relative price ratio P_t/P_{t-1} is greater than 1 then returns are positive and if it is less than 1 returns are negative.

$R_t = 80\%$ wow!

0%

Log Returns

$\exp(r_t) = \exp(\log(1+R_t))$

$R_t = \exp(r_t) - 1$

given

Log Returns r_t

The log return of an asset is defined as

$$r_t = \log(1 + R_t) = \log P_t - \log P_{t-1}$$

slide 4

Log returns are also referred to as continuously compounded (cc) returns.

log

$\updownarrow$

$\ln(\exp(x)) = x$

Prices, simple returns, and log returns

The next slide shows the monthly prices for Microsoft (msft) stock (column 1), simple returns (column 2), and log returns (column 3).

Note that no return figures are reported for January 1993. Their absence emphasises that an observation is lost at the beginning of the sample when computing returns because the price of the stock before the start of the sample period is not available.

$ R_t ≈ r_t

	MSFT	SIMPLE_RETURN	LOG_RETURN
1993M01	1.92	NA	NA
1993M02	1.85	-0.036458	-0.037140
1993M03	2.06	0.113514 >0	0.107520
1993M04	1.90	-0.077670	-0.080852
1993M05	2.06	0.084211	0.080852
1993M06	1.96	-0.048544	-0.049762
1993M07	1.65	-0.158163	-0.172169
1993M08	1.67	0.012121	0.012048
1993M09	1.83	0.095808	0.091492
1993M10	1.78	-0.027322	-0.027703
1993M11	1.78	0.000000	0.000000
1993M12	1.79	0.005618	0.005602
1994M01	1.89	0.055866	0.054361
1994M02	1.83	-0.031746	-0.032261

$$1993\,M3 = \frac{2.06 - 1.85}{1.85} = -0.11$$

Simple and log returns

The simple and log returns to Microsoft for February 1993 are respectively

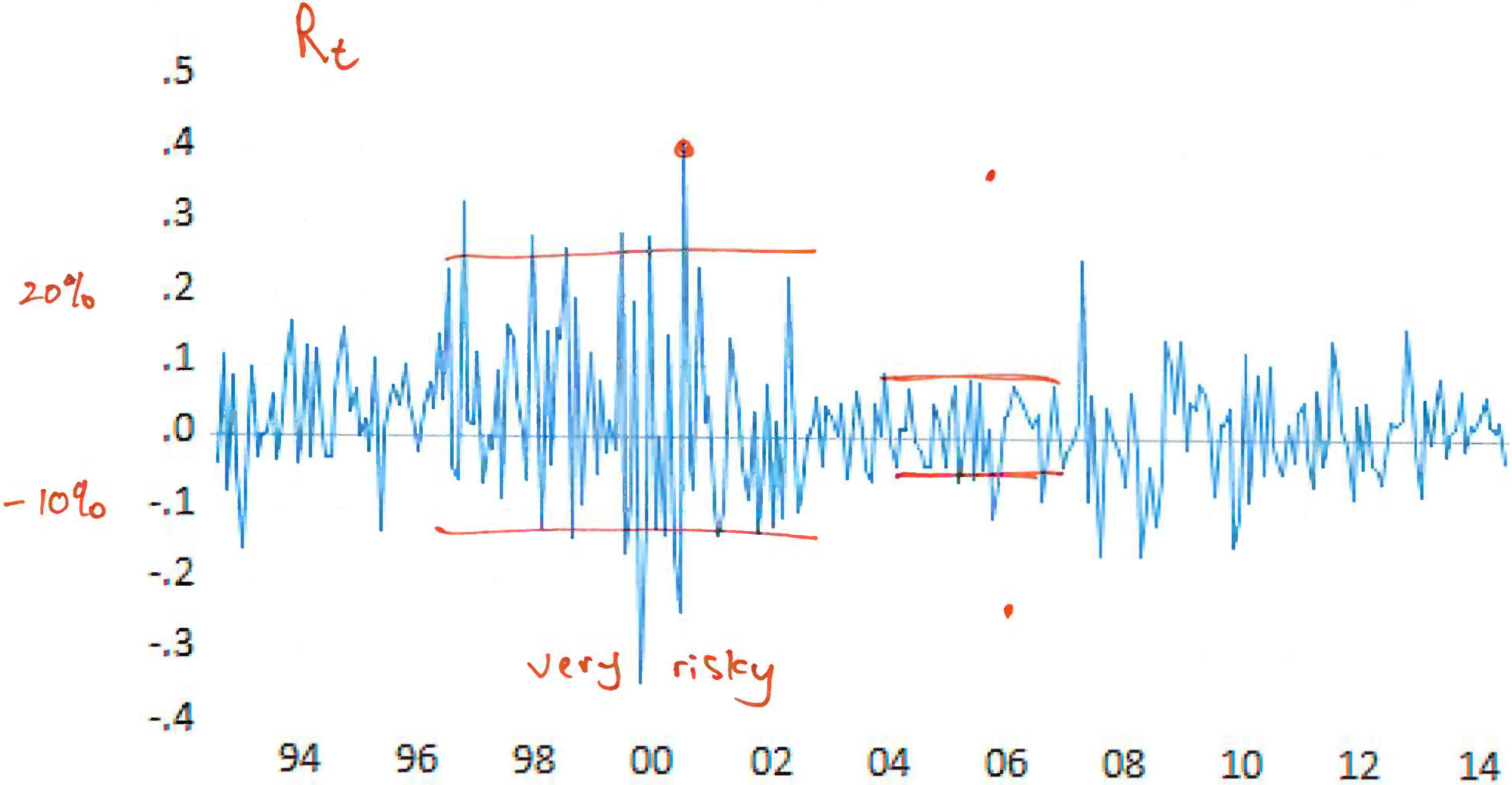
$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{1.85 - 1.92}{1.92} = -0.03645833$$

and

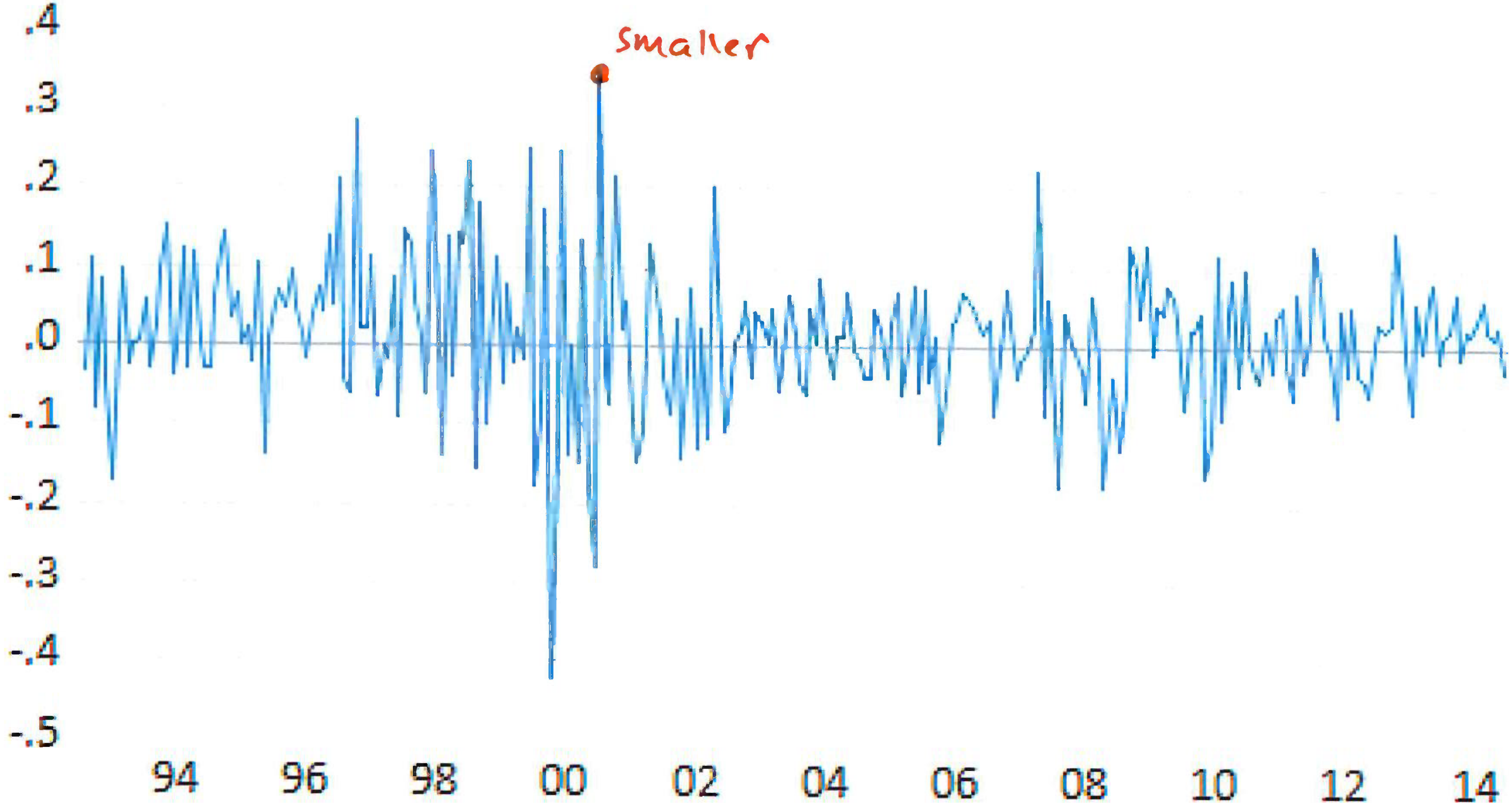
given $$r_t = \log P_t - \log P_{t-1} = \log(1.85) - \log(1.92) = -0.03713955$$

$$R_t = \exp(r_t) - 1 = \exp(-0.0371\ldots) - 1 =$$

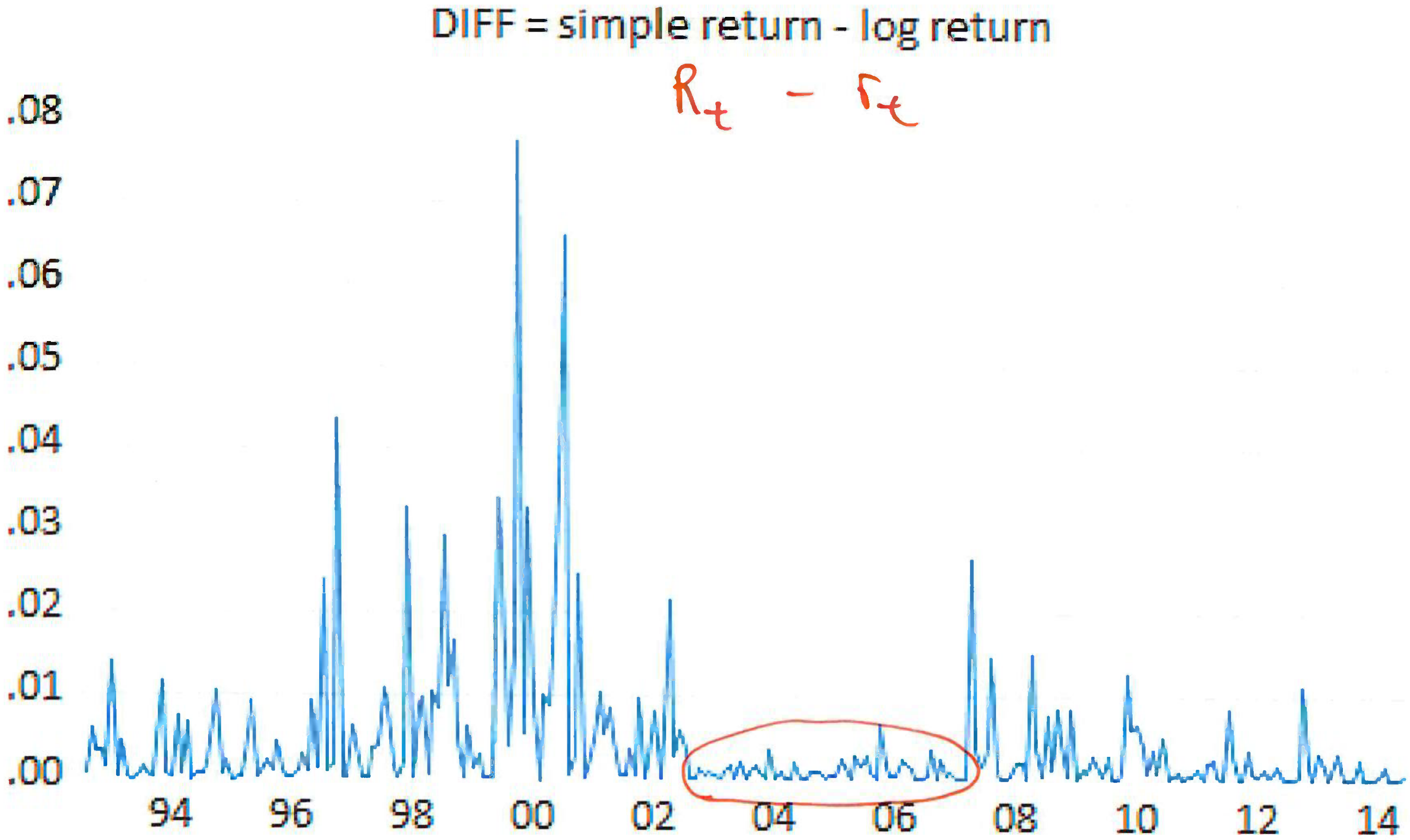
Simple returns on Microsoft (Jan 1993 to Dec 2014)



Log returns on Microsoft (Jan 1993 to Dec 2014)



Simple returns minus log returns



Simple returns and log returns

The previous slide illustrates the difference between simple and log returns.

Qualitatively, the simple and cc returns (or log returns) look almost identical.

The main differences occur when the simple returns are large in absolute value (e.g., mid 2000, early 2001 and early 2008).

Stock Market Indices

$\underbrace{\text{market index}}_{\text{the market}} \Rightarrow$ "benchmark"

A problem of particular importance is the return to a portfolio that comprises all or at least a selection of prominent stocks on a stock exchange.

Stock market index is an aggregate summary measure of the performance of the stock market as a whole.

For example, the S&P 500 (Standard and Poor's 500) is a stock market index that tracks the performance of 500 of the largest publicly traded companies in the United States.

It is widely regarded as one of the best indicators of overall stock market performance and economic health.

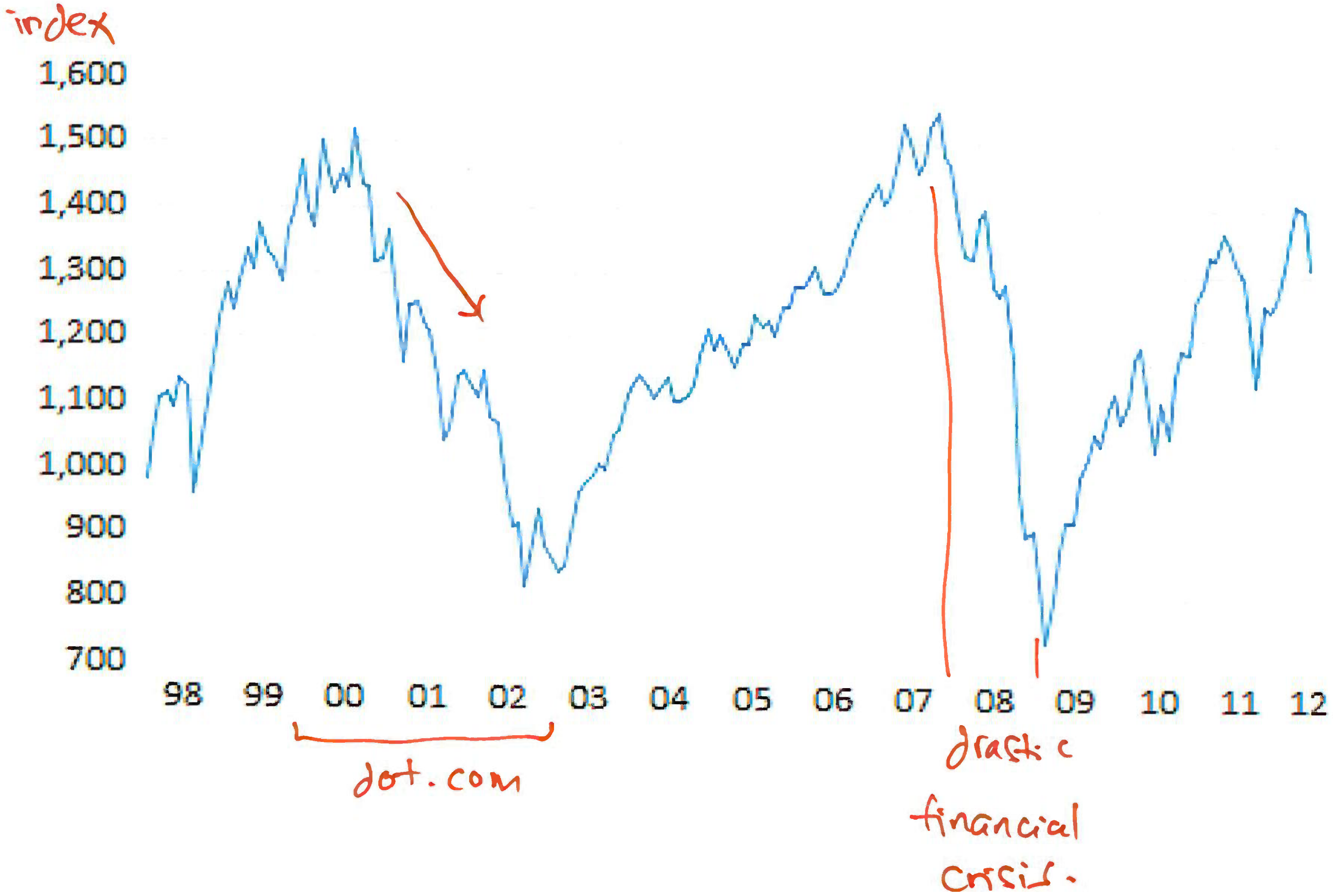
Stock Market Indices

Indices combine a selection of (a large number of) stocks in a particular way to create a portfolio.

The index then represents the value of the portfolio and is expressed in terms of an average price that has been normalised in some way.

Because stock market indices are price indices, the computation of returns to the index can be performed exactly the same way as if it were a single stock.

S&P 500 index from January 1998 to May 2012.



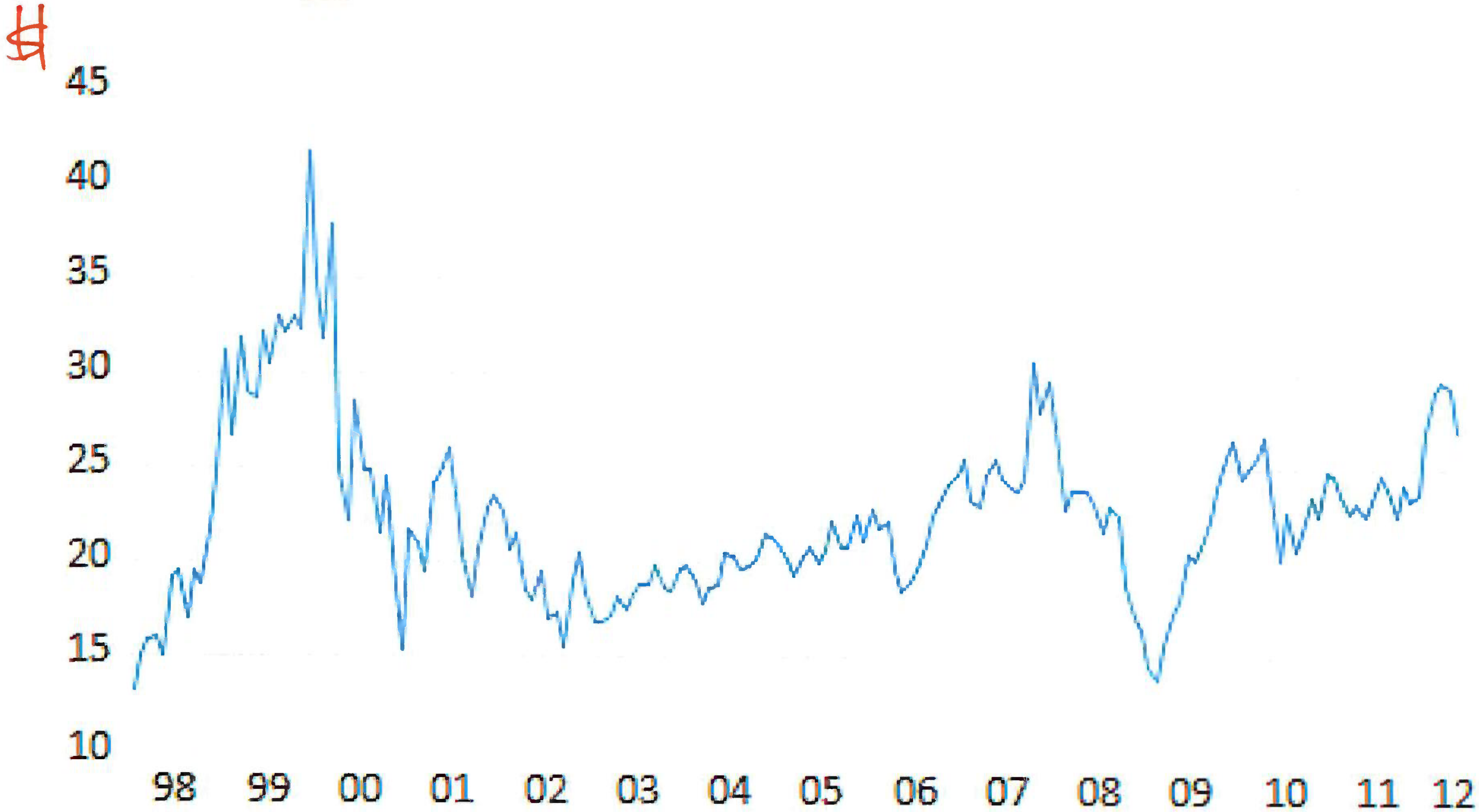
S&P 500

The falls in S&P 500 that occurred around the collapse of the dotcom bubble in the early 2000s and the global financial crisis of 2008-2009 are evident in the graph.

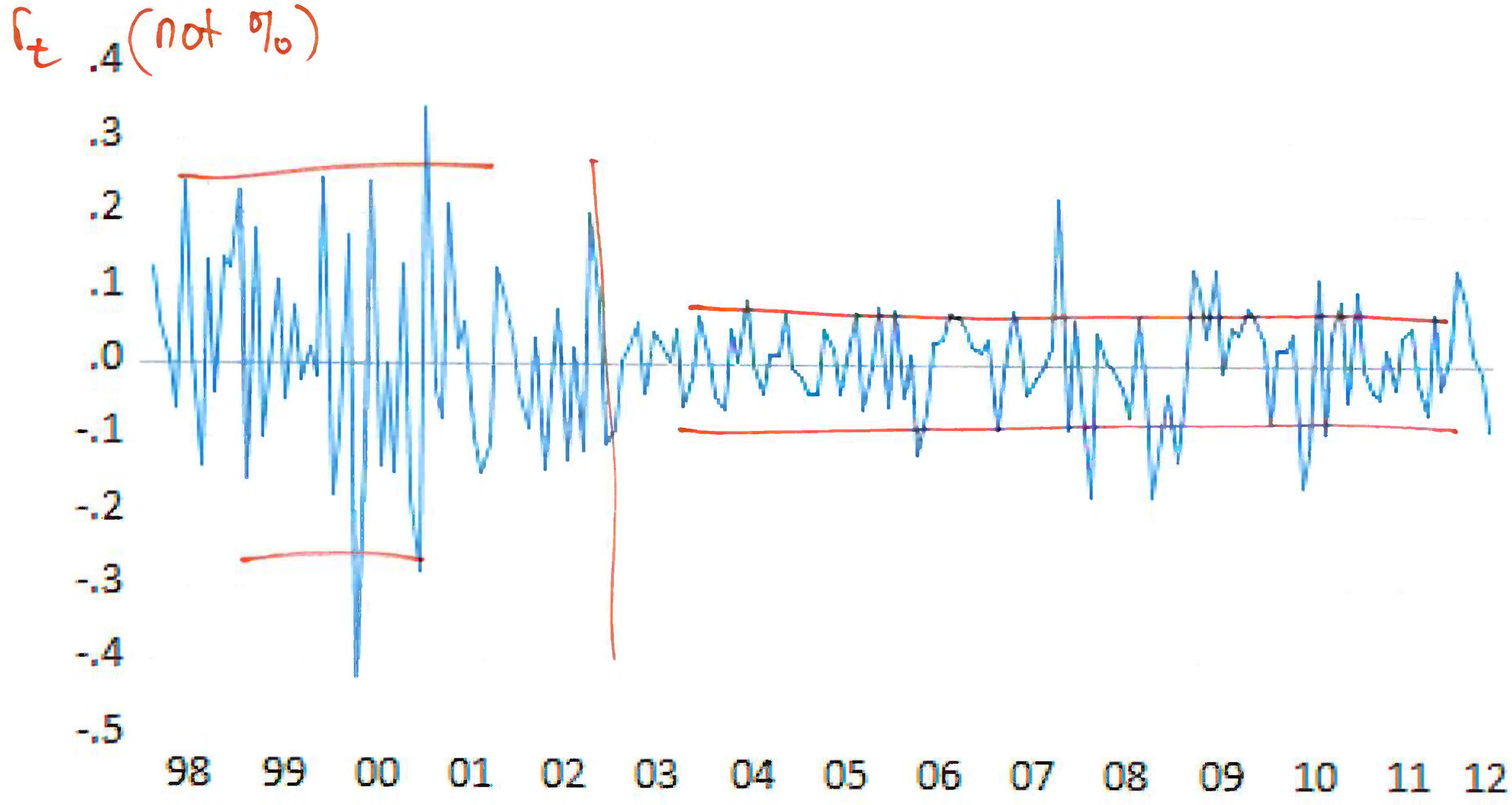
The next slide plots Microsoft stock from January 1998 to May 2012.

Both prices show two large boom-bust periods associated with the dot-com period of the late 1990s and the run-up to the financial crisis of 2008.

Microsoft stock from January 1998 to May 2012.



Monthly log returns on Microsoft stock from Jan 1998 to May 2012.

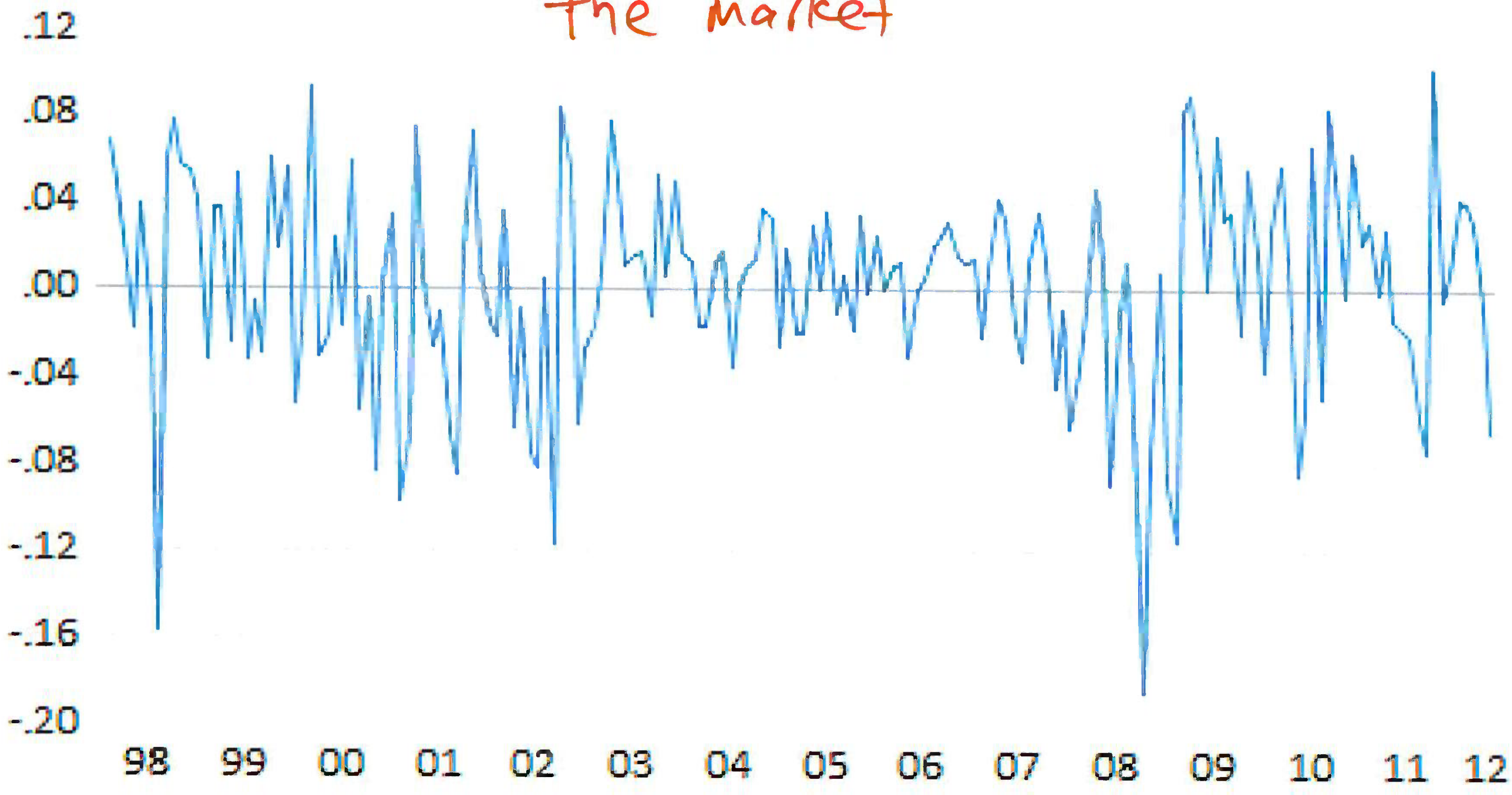


usually use log return

seldom use simple return.

Monthly log returns on S&P 500 index from Jan 1998 to May 2012.

the "market"



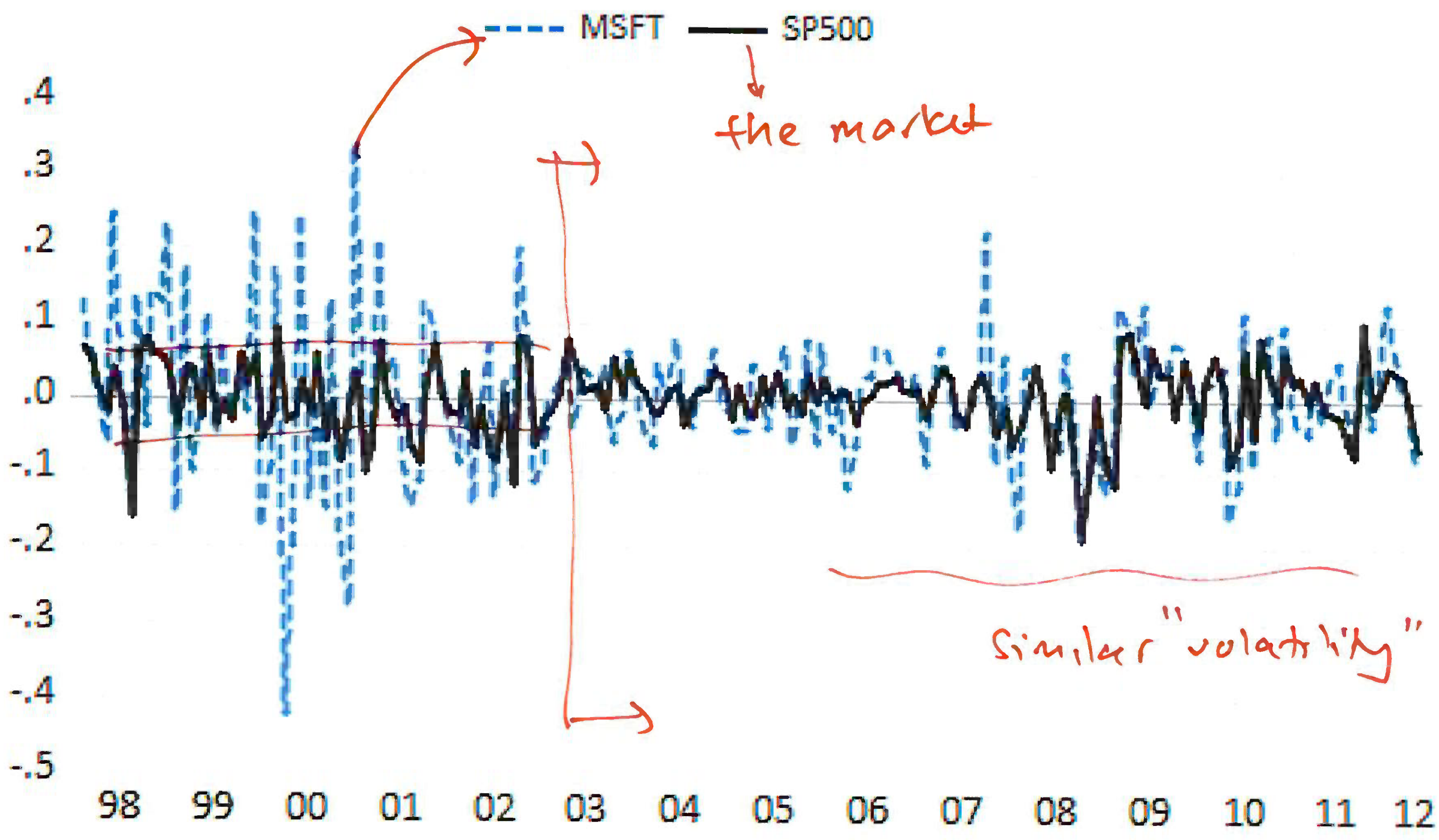
Log returns on Microsoft and the S&P 500

The volatility of the monthly returns on Microsoft and the S&P 500 looks to be similar but this is illusory.

The y-axis scale for Microsoft is much larger than the scale for the S&P 500 index and so the volatility of Microsoft returns is actually much larger than the volatility of the S&P 500 returns.

The next slide shows both return series on the same time plot.

Monthly log returns on Microsoft stock and the S&P 500 index.



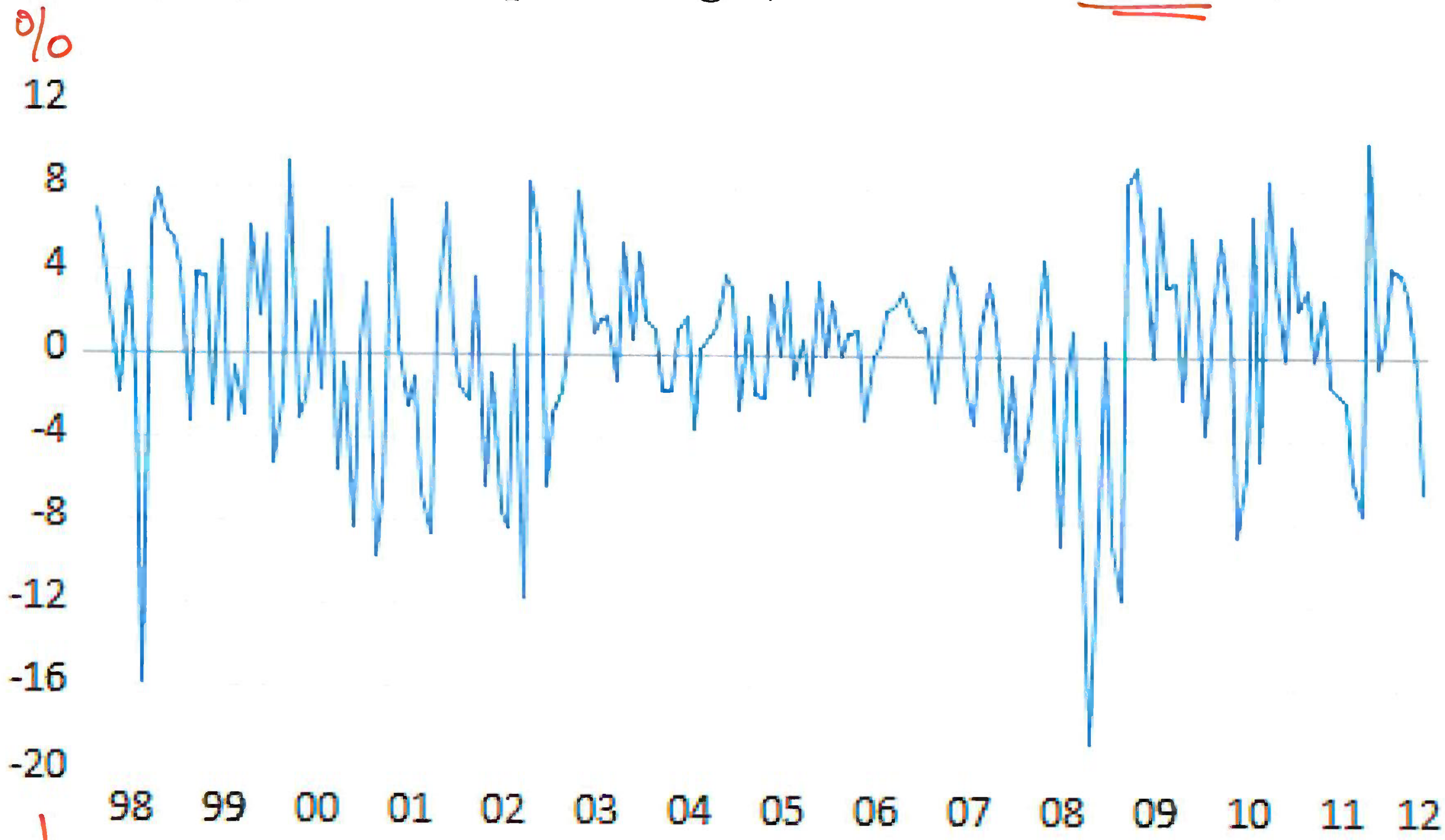
Log returns on Microsoft and the S&P 500

Now the higher volatility of Microsoft returns, especially before 2003, is clearly visible.

However, after 2008 the volatilities of the two series look quite similar.

In general, the lower volatility of the S&P 500 index represents risk reduction due to holding a large diversified portfolio.

Monthly log returns (**in percentages**) for the S&P 500 index.



not 0.2

0.4

− 0.8

0. something

Excess Return

$r_{ft} \Rightarrow$ risk free rate.

The difference between the return on a risky financial asset and a risk free interest rate, denoted r_{ft}, is known as the excess return. The risk free rate is often taken to be the interest rate on a government bond.

The simple excess return (or excess simple return) on an asset is defined as

$$R_t - r_{ft}$$

$\hookrightarrow$ risk free

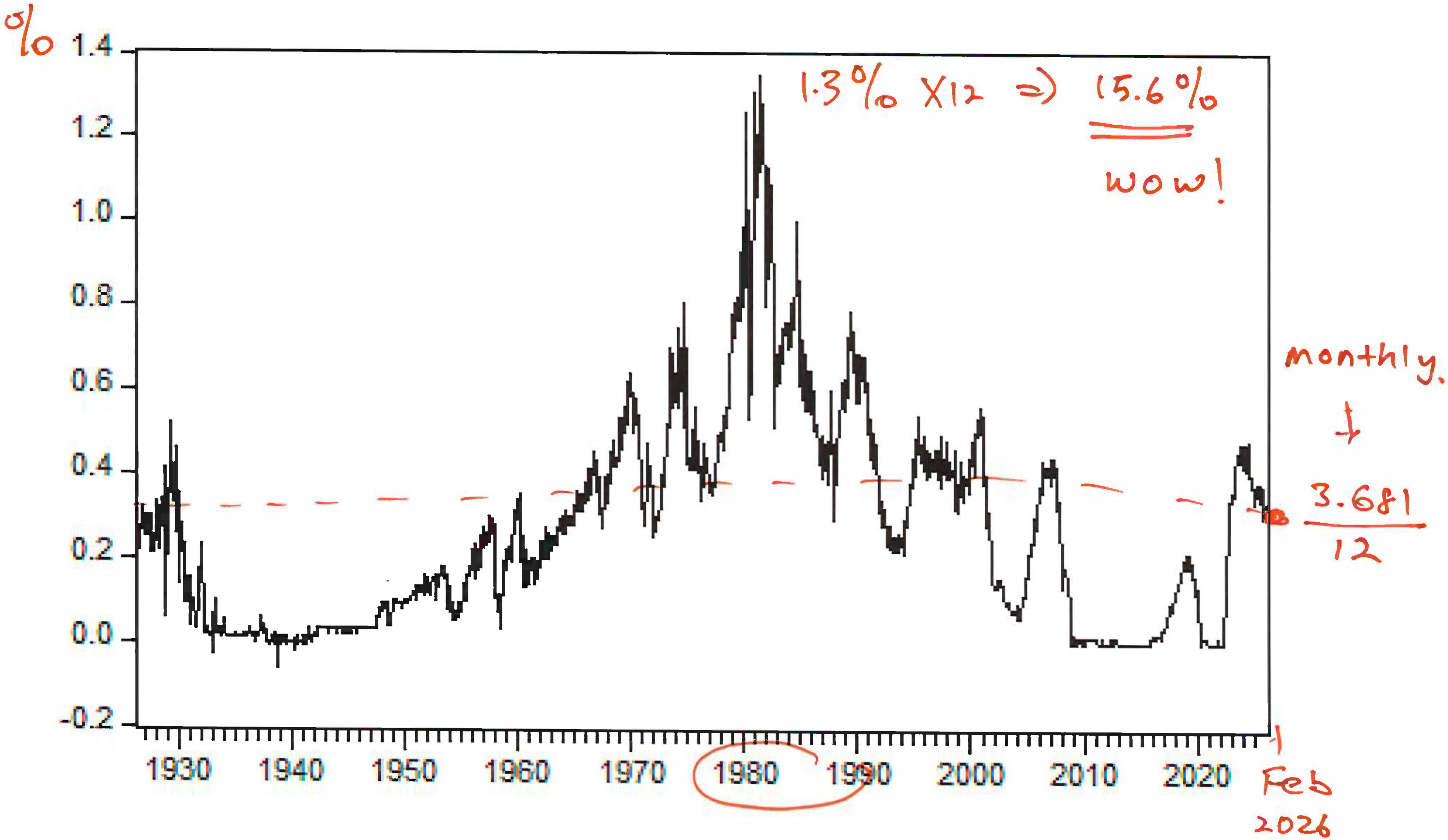
The log excess return (or excess log return) on an asset is defined as

$$r_t - r_{ft}$$

Typically, the 1-month Treasury bill (T-bill) rate is used as a proxy for the risk free rate.

risk free

This figure plots the monthly 1-month T-bill rate (%) (or monthly risk free rate, r_{ft}) from July 1926 to December 2024.



U.S. Risk Free rate

U.S. 1 Month Treasury Bill

3.681% p.a.

0.00

Last Updated: Feb 27, 2026 8:55 a.m. EST

Australia Risk Free rate

RESERVE BANK OF AUSTRALIA

Cash rate target

3.85% p.a.

Effective date 4 February 2026
Next update 2.30 pm, 17 March 2026

Inflation

3.8% CONSUMER PRICE INDEX
Annual change January month 2026

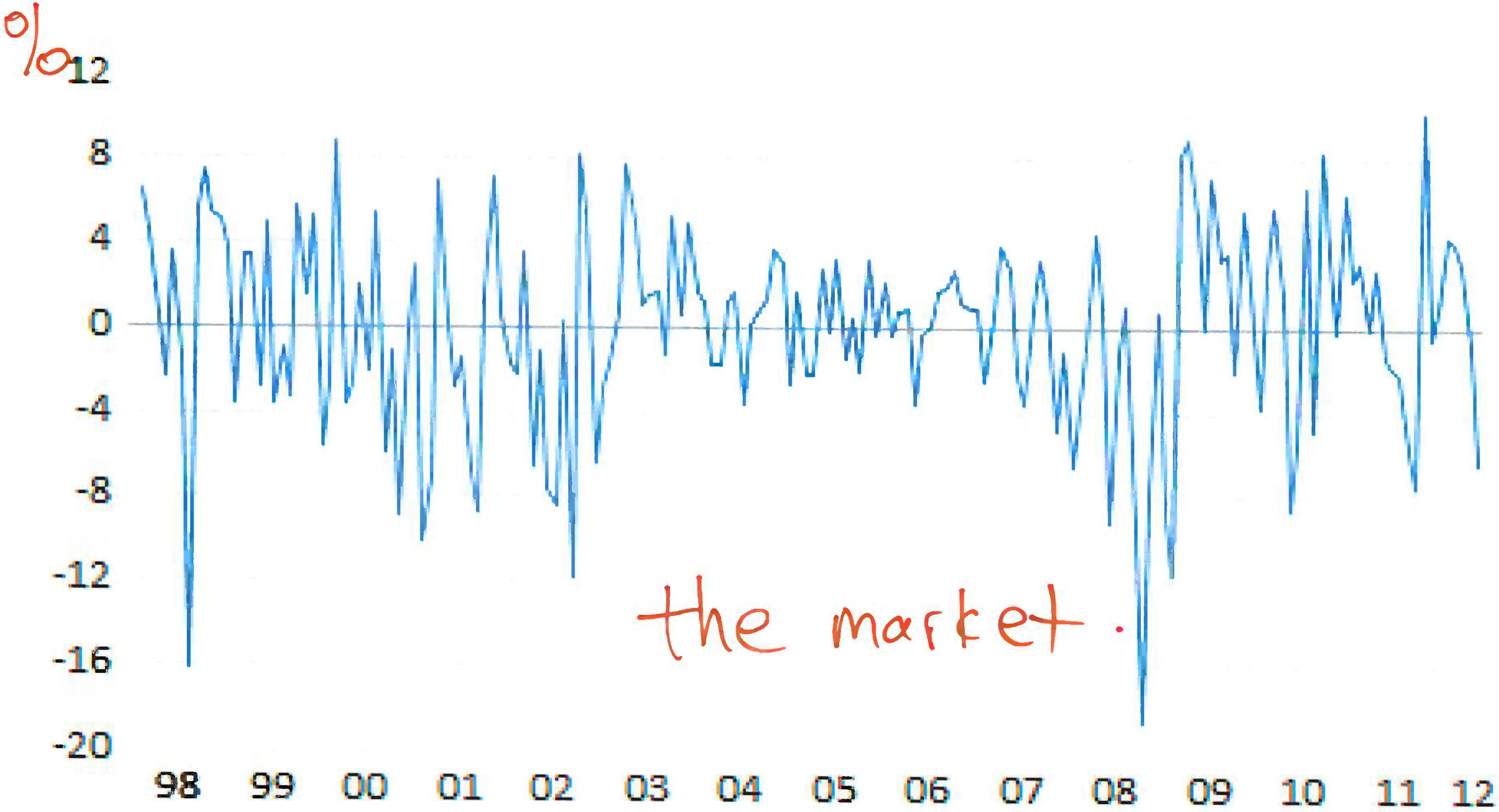
Next update 25 March 2026

S&P 500 log returns, risk-free, log excess return

monthly r_t − monthly r_{ft} = excess return.

	SP500_LR_P	RF	LOG_EXCESS_RET
2011M08	-5.846749%	→ 0.01%	-5.856749
2011M09	-7.446713	0.00	-7.446713
2011M10	10.23066	0.00	10.23066
2011M11	-0.507148	0.00	-0.507148
2011M12	0.849655	0.00	0.849655
2012M01	4.266000	0.00	4.266000
2012M02	3.978733	0.00	3.978733
2012M03	3.085154	0.00	3.085154
2012M04	-0.752574	0.00	-0.752574
2012M05	-6.469925	0.01	-6.479925

This figure plots monthly log **excess returns** on the S&P 500 Index from January 1998 to May 2012.



$r_t - r_{f,t}$

% month   % month.

1990 Nobel Price, William F. Sharpe (Week 5)

William F. Sharpe

Facts

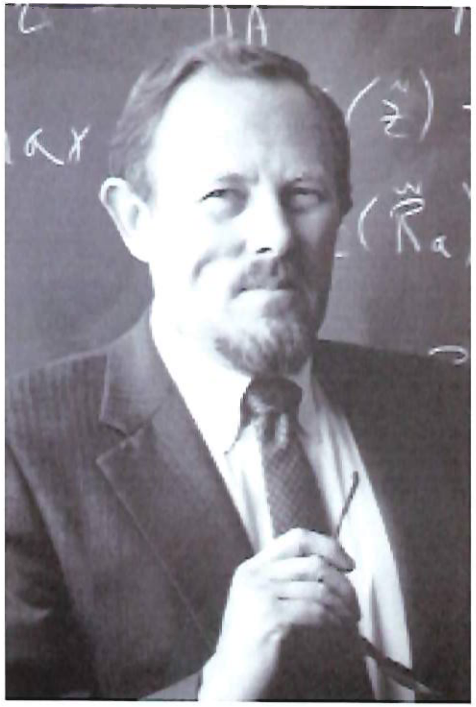


Photo from the Nobel Foundation archive.

William F. Sharpe

Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 1990

Born: 16 June 1934, Boston, MA, USA

Affiliation at the time of the award: Stanford University, Stanford, CA, USA

Prize motivation: "for their pioneering work in the theory of financial economics"

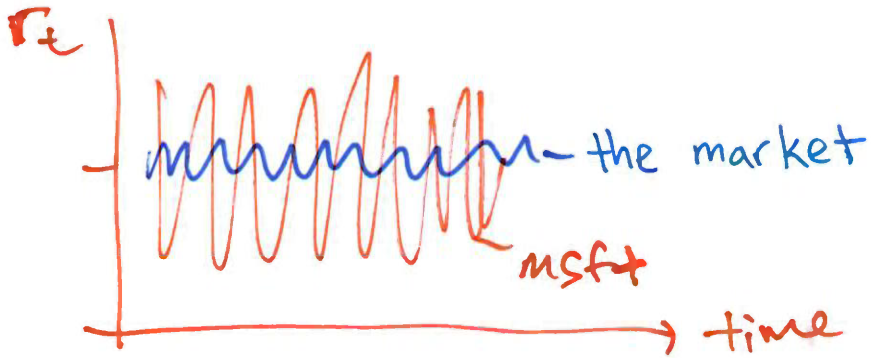
Market Risk (Week 5)

Market risk: If the market *excess return* changes by 1%, the asset's A *excess return* changes by about 1.5% *on average*.

Week 5 model.

- When the market return **rises** by 1 per cent, the asset return is expected to rise by about 1.5 per cent on average.
- When the market return **falls** by 1 per cent, the asset return is expected to fall by about 1.5 per cent on average.

This A asset is riskier than the market, and so it is expected to command a higher expected return than the market portfolio.



Asset A is riskier than the market

higher return on average

Market risk premium

$E(R_t) \Rightarrow 8\%$ $\overset{\text{minus}}{-}$ $r_f \Rightarrow 7\%$

average R_t over a long time.

1%

Market risk premium is defined as the expected return of the market portfolio in excess of the risk free rate:

$$\text{Market risk premium} = \mathbb{E}[R_m] - r_f. \quad > 0 \quad = 1\%$$

Here, $\mathbb{E}[R_m]$ is the expected return on the market portfolio.

Market excess return means the realised market return minus the risk-free rate:

$$\text{Market excess return at time } t = R_{m,t} - r_{ft}.$$

2013 Nobel, Eugene F. Fama (Week 5)

The Father of Modern Empirical Finance

The University of Chicago Booth School of Business

5807 S. Woodlawn Ave.
Chicago, IL 60637 USA

why?

David G. Booth, co-founder of Dimensional Fund Advisors

From Theory to Trillions: David Booth | Financial Thought Exchange

By Cathy Scott

CFA Institute

Wikipedia - David G. Booth

Wikipedia website
Booth has spent his career applying groundbreaking financial theory and research to the world of asset management. His investment approach with Dimensional is based on the application of the efficient market hypothesis, and he helped pioneer what would later be called factor investing, targeting areas of the market that research shows have higher expected returns.

Dimensional's researchers and academics seek to develop investment strategies that will outperform index funds.

Quant (factor) Investing

In quantitative investing, investors use measurable stock characteristics as signals.

Many quantitative investors believe these signals serve as proxies for underlying fundamentals that help explain expected returns.

higher return ⇒ some characteristics

e.g. ~~age~~

~~gender~~

1:00

1:05 back.

Popular quant signals

Here are two widely used categories of quant signals.

age ⇒ (a) Size: Market capitalisation (market cap) is defined as

$$\text{Market cap} = \underset{\text{up / down}}{\text{share price}} \times \text{number of shares outstanding.}$$

(b) Valuation: How is the company priced relative to fundamental accounting measure? For this, we have the widely used book-to-market (BtM) ratio:

accountant → Assets − liabilities

$$\text{BtM} = \frac{\text{book value of "equity"}}{\text{market value of equity}}$$

"investor"

$$BtM = \frac{1}{yahoo}$$

yahoo ⇒ Price / book

Big stocks vs Small stocks in Fama-French terms

Market Cap
Example: If a company's share price is $1 and it has 100 million shares outstanding, then its market cap is $100 million.

In the Fama–French framework:

- "Big" stocks typically refer to large-cap or mega-cap firms (household names).
- "Small" stocks typically refer to small-cap firms (i.e, they are smaller, less widely owned).

Classic Big stocks

- Apple
- Microsoft

AAPL = 3.89 T
MSFT

Small stocks

SOS ⇒ 11.05 m

less risky than
lower return than

Value stocks vs Growth stocks in Fama–French terms

Value stocks (high BtM)

A value stock is a stock with a high book-to-market ratio (high BtM).

A high BtM often indicates the stock is priced relatively low compared with its accounting book value (sometimes reflecting financial distress, low investor sentiment, or limited investor attention).

General Motors (GM)

2006 $\Rightarrow$ BtM $= 1.28$ $\Rightarrow$ MtB $= \frac{1}{1.28} = 0.78$ dollar for \$1 in book value

⋮

1 June 2009 $\Rightarrow$ bankruptcy protection $\Rightarrow$ in general very high BtM

⋮

now $\rightarrow$ BtM $= \frac{1}{1.15} = 0.87$

Price/Book = yahoo finance

Value stocks vs Growth stocks in Fama–French terms

Growth stocks (low BtM)

A growth stock is a stock with a low book-to-market ratio (low BtM).

These are stocks the market prices highly relative to book value, often because investors expect strong future growth.

google

2006 $\Rightarrow$ BtM = 0.04

⋮

now $\Rightarrow$ BtM = 0.112 $\Rightarrow \frac{1}{\text{yahoo}} = \frac{1}{8.92}$

less risky.

Example 1 (low BtM: growth-like).

Fama

Suppose Stock A has BtM = 0.095. Then, for each \$1 of market value, the book value is \$0.095.

The market-to-book ratio is

$$\frac{1}{0.095} = 10.53.$$

That is, for each \$1 of book value, the market value is about \$10.53.

Investors value this stock to the extent that they are willing to pay much more than the existing book value.

This is characteristic of growth-style stocks – low BtM.

Example 2 (high BtM: value-like).

more risky

Suppose Stock B has BtM $= 1.339$. Then, for each \$1 of market value, the book value is \$1.339.

The market-to-book ratio is

$$\frac{1}{1.339} = 0.75.$$

That is, for each \$1 of book value, the market value is \$0.75.

Basically, investors are not willing to pay the full book value for the stock.

This is characteristic of value-style stocks – high BtM.

Value premium and Size premium

P/E low ≈ P/E high

Historically in US data, value stocks have earned higher average returns than growth stocks (the "value premium").

Also, small stocks have earned higher average returns than large stocks (the "size premium").

A

B

* size ← ① small stocks

sos

> ① Big stocks

msft

* value ← ② value stocks

② growth stocks

more risky than

higher return than on average.

Value at Risk

Value at Risk (VaR) is a single estimate of the amount by which an institution's position in a risk category could decline due to general market movements during a given holding period.

The measure can be used by financial institutions to assess their risks or by a regulatory committee to set margin requirements.

In either case, VaR is used to ensure that the financial institutions can still be in business after a catastrophic event.

Quantiles

Quantiles from a general normal distribution.

Let R denote the simple return and assume that

$$R \sim N\left(\mu, \sigma^2\right).$$

Quantiles of R (denoted as q_α) can be computed using

$$q_\alpha = \mu + \sigma\, z_\alpha.$$

Standardizing a Random Variable

Let X be a random variable with $E(X) = \mu$ and $var(X) = \sigma^2$.

Define a new random variable Z as

$$Z = \frac{X - \mu}{\sigma},$$

This transformation is called *standardizing* the random variable X and is often called a z-score.

Standardization creates a new random variable Z with mean zero and variance 1.

Standardizing a Random Variable

In addition, if

$$X \sim N(\mu, \sigma^2)$$

is normally distributed then

$$Z \sim N(0, 1).$$

VaR Example

Consider an investment of \$10,000 in Microsoft stock over the next month. Let R denote the monthly simple return on Microsoft stock and assume that

$$R \sim N\left(0.05,\ (0.10)^2\right).$$

That is, $E(R) = \mu_R = 0.05$ and $\mathrm{var}(R) = \sigma_R^2 = (0.10)^2$.

Let W_0 denote the investment value at the beginning of the month and W_1 denote the investment value at the end of the month.

In this example, $W_0 = \$10{,}000$.

VaR Example

Consider the following questions:

(i) What is the probability distribution of end-of-month wealth W_1?

(ii) What is the loss in dollars that would occur if the return on Microsoft stock is equal to its 5% quantile, $q_{0.05}$? That is, what is the monthly 5% VaR on the \$10,000 investment in Microsoft?

(i) End of month wealth W_1

Fey $$W_1 = W_0(1+R) = W_0 + W_0 R = \$10000 + \$10000R$$

and so

$$E(W_1) = E(W_0) + W_0 E(R) = \$10,000 + \$10,000(0.05) = \$10,500$$

and

$$\begin{aligned} var(W_1) &= (W_0)^2\, var(R) = (\$10,000)^2(0.10)^2 \\ sd(W_1) &= (\$10,000)(0.10) = \$1000 \end{aligned}$$

Since R is assumed to be normally distributed it follows that W_1 is normally distributed:

$$W_1 \sim N\left(\$10,500, (\$1000)^2\right).$$

(ii) Since $R \sim N\left(0.05, (0.10)^2\right)$ we solve for the 5% quantile of Microsoft stock

$$q_{0.05} = \mu_R + \sigma_R \cdot z_{0.05} = 0.05 + 0.10\,(-1.645) = -0.1145$$

With 5% probability the return on Microsoft stock is -11.45% or less.

Now, if the return on Microsoft stock is -11.45% the loss in investment value is $(\$10,000)\,(0.1145) = \$1145.$

Hence, \$1,145 is the 5% VaR over the next month on the \$10,000 investment in Microsoft stock.

Summary - VaR

If $R \sim N\left(\mu, \sigma^2\right)$, the VaR_α can be calculated as follows:

Step 1: Calculate the quantile q_α as

$$q_\alpha = \mu + \sigma \, z_\alpha$$

Step 2: Calculate the (next period) $\alpha \times 100\%$ VaR as

$$VaR_\alpha = |W_0 \times q_\alpha|$$

where W_0 denote the initial investment.

In words, VaR_α represents the dollar loss that could occur with probability α. By convention, it is reported as a positive number (hence the use of the absolute value function).

Summary - VaR

Step 1:

Calculate the $q_{0.05}$ as

$$\begin{aligned} q_{0.05} &= \mu + \sigma\, z_{0.05} \\ &= 0.05 + 0.1\,(-1.645) \\ &= -0.1145 \end{aligned}$$

Step 2:

Compute the 5% VaR values over the next month

$$\begin{aligned} \mathrm{VaR}_{0.05} &= |W_0 \times q_\alpha| \\ &= |10000 \times (-0.1145)| \\ &= \$1145 \end{aligned}$$

Hence with 5% probability the loss over the next month is at least \$1,145.

VaR

slide 8 $\Rightarrow R_t = \exp(r_t) - 1$

When dealing with log returns r_t, if necessary, one can also use the approximation

$$VaR = |W_0 \cdot (\exp(q_\alpha^r) - 1)|$$

Don't do this

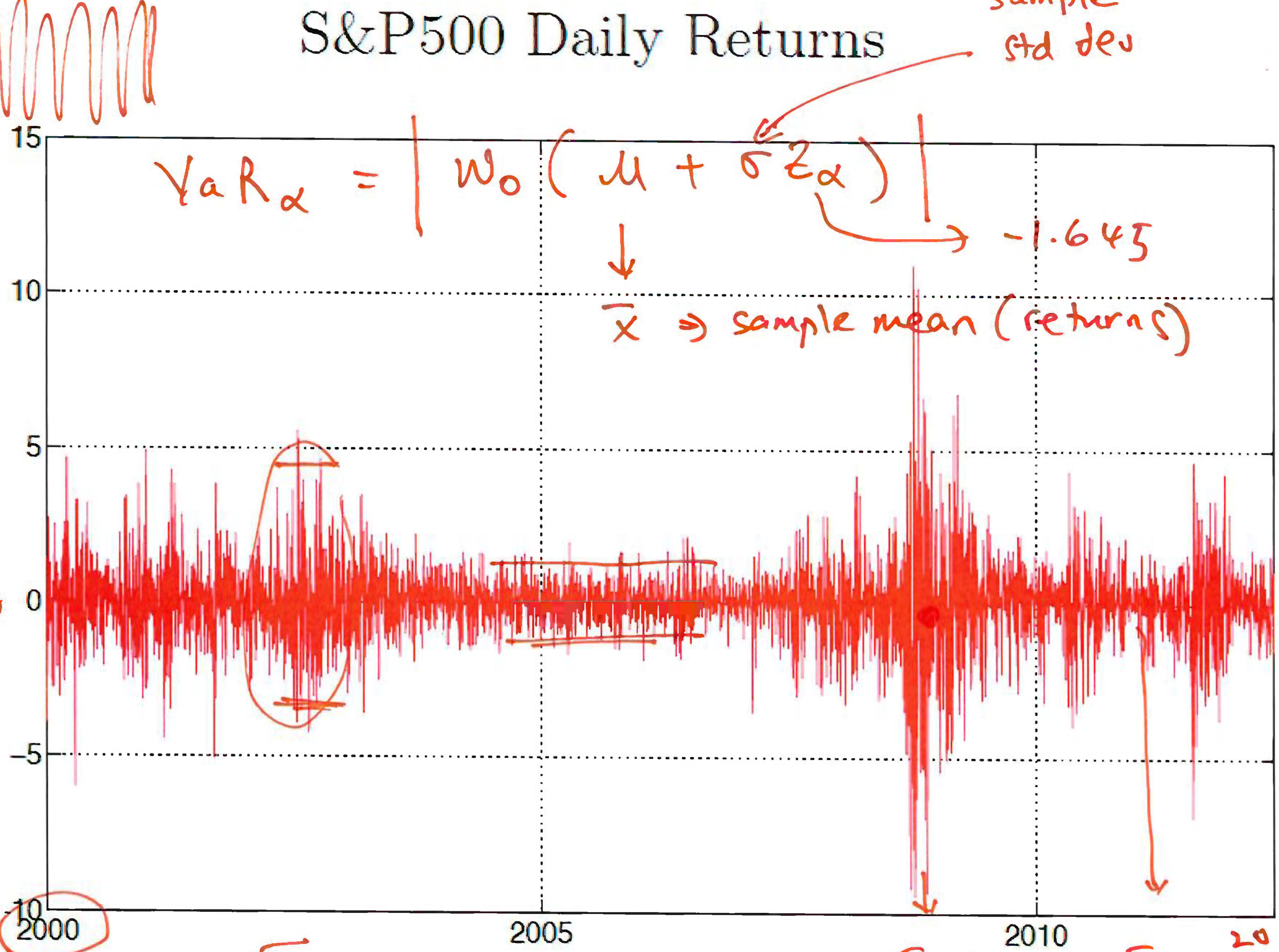
where q_α^r is the $\alpha \times 100\%$ quantile from the normal distribution for the log return.

Practitioners often omit this approximation due to its negligible effect.

Since log returns correspond approximately to simple returns of a financial asset, we thus use log returns.

Constant Variance Over Time – What do you think?

constant variance over time



sample std dev

$$VaR_\alpha = \left| w_0 \left(\mu + \sigma z_\alpha \right) \right|$$

$z_\alpha \rightarrow -1.645$

$\mu \rightarrow \bar{x} \Rightarrow$ sample mean (returns)

σ_t

$\sigma_t > \sigma_t$

20

JPMorgan Chase & Co. is the world's largest bank by market capitalization (as of 2025) and the largest in the U.S., with over $4 trillion in assets. As a global financial leader, it provides investment banking, asset management, and private banking services to governments, corporations, and institutions in over 100 countries. J.P. Morgan +3

J.P. Morgan: The Banker Who Shaped Modern Wall Street

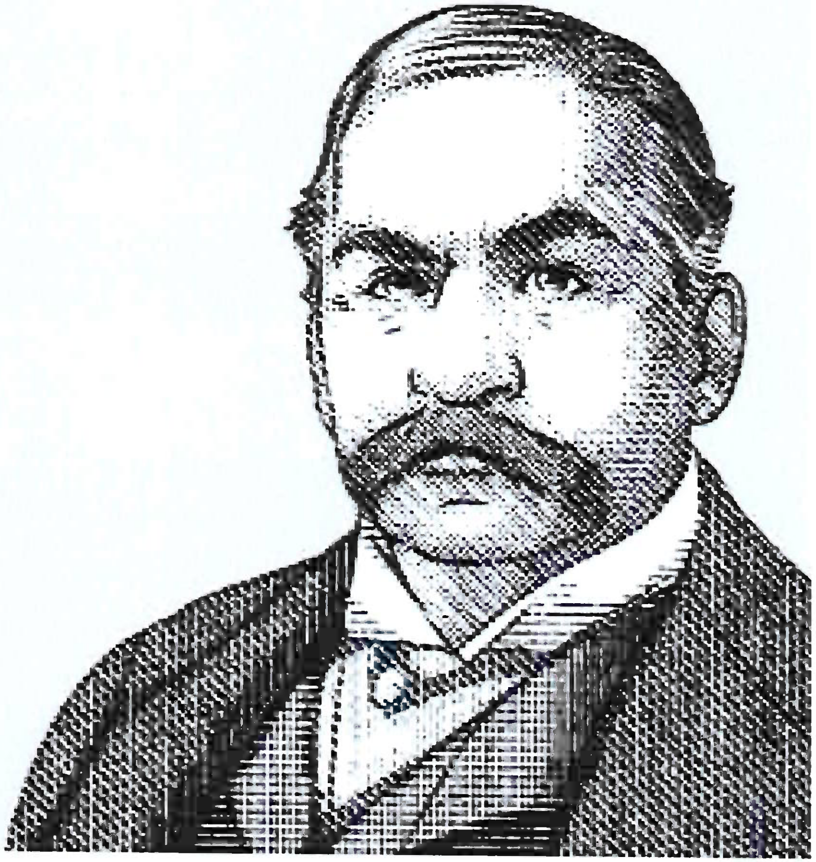
J.P. Morgan

Born: April 17, 1837

Died: March 31, 1913

Financier and Banker

- Helped lead the creation of the Federal Reserve System
- Used his personal power and reputation to encourage the formation of trusts and mergers within industries where he saw competition
- Financed industrial consolidations that created General Electric



J. P. Morgan RiskMetrics (Week 11)

J. P. Morgan developed the RiskMetrics method to VaR calculation.

RiskMetrics assumes time-varying variances σ_t:

What?

$$\sigma_t^2 = (1 - \lambda) \sum_{j=0}^{\infty} \lambda^j r_{t-j-1}^2$$

where r_t is the demeaned log return, and $\lambda = 0.94$.